

Chapter 161 — Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

IF YOU READ NOTHING ELSE

- **Absolute contraindications:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Robotic vs open (Yaxley 2016 randomised controlled trial [RCT]):** equal oncology/function; robot ↓ blood loss (300 vs 1400 mL), ↓ stay.
- **Surgeon volume/experience,** not platform, drives margins and function.
- **Interfascial plane** = nerve-sparing default for organ-confined disease.
- **Apex** = commonest positive-margin site.
- **Avoid thermal energy** near neurovascular bundles (NVBs); clips + cold cut on pedicles.
- **Catheter ~1 week;** not before day 5 (retention ↑).
- **Bladder neck stricture** <2% robotic vs up to 16% open.
- **Pelvic lymph node dissection (PLND):** if done → extended template.
- **Lymphocele** = commonest PLND complication; transperitoneal not protective.

EVOLUTION

- **Walsh and Donker 1982:** NVB anatomy → nerve-sparing surgery.
- **First laparoscopic radical prostatectomy (LRP):** Schuessler 1997 — no advantage shown, not adopted.
- **da Vinci:** 2000 launch; S 2006 (4th arm, HD); Si 2009 (dual console); Xi 2014 (side docking); SP 2018 (single port).
- **View:** 10× magnified, three-dimensional; wristed instruments.
- **Robotic-assisted (RALRP)** = dominant US approach; LRP where robots scarce.

PATIENT SELECTION

- **Indications** = open surgery: cT1–T2, or resectable cT3.
- **High risk** → pelvic computed tomography (CT) or magnetic resonance imaging (MRI) + bone scan; unfavourable intermediate → consider.
- **Prostate-specific membrane antigen positron emission tomography (PSMA PET)-CT:** high risk with negative conventional imaging.
- **Relative only:** obesity, prostate >80 g, median lobe, prior transurethral resection of the prostate (TURP), neoadjuvant hormones, prior pelvic surgery, salvage.
- **Prior extraperitoneal mesh repair** → transperitoneal.
- **Obesity:** 16-cm trocars; ports 1–2 cm cephalad; pressure ↓ 10 mm Hg.
- **Multiparametric MRI (mpMRI) staging:** specificity 0.88, sensitivity 0.61.
- **Large gland:** ↑ time/blood loss; continence unchanged.

PREOPERATIVE PREPARATION

- **Antibiotic:** cefazolin pre-incision; positive culture → targeted.
- **Low-dose aspirin:** continue.
- **General anaesthesia mandatory** — never regional.
- **Baseline:** Sexual Health Inventory for Men (SHIM) + International Prostate Symptom Score (IPSS).
- **Fixed shoulder rests** → brachial plexus injury; avoid.
- **Hip hyperflexion** → femoral neuropraxia; Xi needs none.
- **Carbon dioxide (CO₂):** vagal response, oliguria, hypercarbia → acidosis.
- **Fluids:** 1.5–2 L crystalloid, mostly after anastomosis.
- **Facial/orbital oedema** → suspect laryngeal oedema; delay extubation.

INSTRUMENTATION

- **Robot arms:** left bipolar grasper, right monopolar scissors, 4th ProGrasp.

- **Anastomosis:** 3-0 Monocryl double-armed; **posterior reconstruction** 2-0.
- **Dorsal venous complex (DVC) suture:** LRP 0 polyglactin vs RALRP 0 polydioxanone.

SURGICAL TECHNIQUE

- **Transperitoneal anterior** = commonest; enter Retzius first.
- **Retrovesical approach:** vesicles first — best for median lobe.
- **Access:** Veress at umbilicus or Hasson; left upper quadrant safe with midline scars.
- **Insufflation:** 12–15 mm Hg; camera 15–17 cm from pubis.
- **Extraperitoneal:** balloon, ~500 mL; ↓ space, ↑ CO₂ absorption; avoids steep tilt.
- **Extra- vs transperitoneal:** no consistent advantage — surgeon preference.
- **Retzius:** incise lateral to medial umbilical ligaments; stay inside pelvic brim.
- **Accessory pudendal arteries:** preserve — may be dominant penile supply.
- **DVC:** ligate as far from apex as possible.
- **Bladder neck cues:** fat line, balloon traction, dome tenting, "pinch".
- **Balloon off midline** → median lobe → divide proximally.
- **Posterior bladder neck:** midline, 45° downward.
- **No bladder neck sparing** after TURP or with median lobe.
- **Denonvilliers incision:** 2–3 cm, ~0.5 cm below vesicle bases.
- **High extracapsular extension (ECE) risk** → excise Denonvilliers (plane on fascia propria).
- **Planes:** extrafascial = widest margin, bilateral → erectile dysfunction (ED); intrafascial → ↑ margins.
- **NVB injury:** incision, entrapment, heat, traction.
- **Cavernous nerves:** 5 and 7 o'clock posterolaterally.
- **Apex:** small urethral rim on apex → ↓ margins, continence intact.
- **Ureteric orifices at bladder neck:** imbricate at 5 and 7 o'clock, or stent.
- **Rocco stitch:** Denonvilliers remnant → posterior rhabdosphincter, 2-0 Monocryl.
- **Van Velthoven:** running anastomosis from 6 o'clock; no eversion sutures.
- **Check haemostasis** at <10 mm Hg; close 12-mm fascia only.
- **Retzius-sparing (Bocciardi 2010):** posterior via pouch of Douglas; anastomosis anterior first.
- **Retzius-sparing:** faster continence to 3 months, equal at 12; potency equal; ↑ positive margins.

- **Hood (Tewari 2021):** anterior approach, preserves detrusor apron/puboprostatic ligaments.

POSTOPERATIVE MANAGEMENT

- **Analgesia:** local infiltration; ketorolac 15–30 mg IV; regular paracetamol + non-steroidal anti-inflammatory drug (NSAID) 24–48 h; opioids last.
- **Diet** within 4–8 h; ileus settles 24–48 h.
- **Same-day discharge:** pressure as low as 6 mm Hg or extraperitoneal.
- **Drain** not routine; out day 1–2.
- **Catheter:** reconstruction/radiation ≥10 days + cystogram; salvage 2–3 weeks.
- **Activity:** normal 4 weeks; strenuous 6 weeks.
- **Pelvic floor exercises:** ≥3 days after catheter removal.
- **ED causes:** neural, vascular, corporal smooth muscle.
- **Memorial Sloan-Kettering (MSKCC) rehabilitation:** phosphodiesterase type 5 inhibitor (PDE5i) failure → intracavernosal injection from week 6.
- **Vacuum device:** preserves penile length.

ONCOLOGIC OUTCOMES

- **Biochemical recurrence (BCR):** prostate-specific antigen (PSA) >0.2 ng/mL after undetectable.
- **Margins robotic = retropubic;** ~15% overall, ~9% pT2.
- **7-year BCR-free** ~80%; cancer-specific survival needs 15–20 years.
- **Margin drivers:** Gleason ≥7, perineural invasion, stage.
- **Fellowship-trained** surgeons outperform open converts.

FUNCTIONAL OUTCOMES

- **Passive continence:** smooth muscle + suspensory system; **active:** striated sphincter.
- **Sphincter:** horseshoe, fast-twitch; pudendal nerve via Alcock's canal.
- **Mild persistent leak** → sling or ProACT; **severe (<2%)** → artificial sphincter.
- **NVB:** sympathetic T12–L1 + parasympathetic S2–S4; 3–5 mm from capsule.
- **At apex** nerves at 3 and 9 o'clock — commonest injury site.
- **Erectile recovery:** up to 5 years; 12-month endpoint too early.
- **Experts:** potency up to 90% at 1 year (no prior ED).

COMPLICATIONS

- **Tilt** 22–25°.
- **Transfusion** 1–2%.
- **Recognised rectal injury, minimal spillage** → multilayer repair + flap, no colostomy.
- **Unrecognised rectal injury** → rectovesical fistula.
- **Ureter at risk:** vas dissection, trigone, proximal PLND.
- **Venous thromboembolism (VTE) prophylaxis (American Urological Association [AUA]):** mechanical for all; heparin only with multiple risk factors.
- **VTE** ~0.5%.
- **Deep vein thrombosis (DVT)** → image pelvis for compressing collection.

- **Stricture:** dilate/incise → urethroplasty.
- **Robot failure** <0.5%; arms and optics.

SALVAGE RALRP

- **Work-up:** biopsy-proven recurrence + PSMA PET.
- **Full bowel preparation;** rectal injury historically >15% (open).
- **Suspected occult rectal injury** → air bubble test.
- **Catheter** 2–3 weeks + cystogram.
- **Continence** 33–80%; usually impotent.

PELVIC LYMPH NODE DISSECTION

- **Templates:** standard = obturator + external iliac; extended adds internal iliac; super-extended adds common iliac + presacral.
- **Capture:** standard 38%, extended 63%, + common iliac 75%.
- **Low risk:** nodes <5% → may omit.
- **AUA 2018:** unfavourable intermediate/high risk; **European Association of Urology (EAU) 2021:** nomogram risk >5%.
- **AUA/American Society for Radiation Oncology (ASTRO) 2022:** risk-stratified, nomogram-based.
- **No survival benefit;** staging only.
- **Extended boundaries:** 2 cm above common iliac bifurcation → node of Cloquet; genitofemoral nerve → bladder.
- **Obturator nerve injury:** medial thigh numbness, weak adduction.
- **Lymphocele:** up to 60% radiological; 2–5% significant.
- **Peritoneal interposition flap** ↓ significant lymphocele.
- **Symptomatic lymphocele** → percutaneous drain → sclerosant/marsupialisation.

FUTURE

- **Key da Vinci patents expired 2019:** Hugo, Versius, Ottava.

CLASSIC TRAPS

- **Trap:** obesity, prior TURP contraindicate RALRP — **Actually:** relative only.
- **Trap:** intrafascial is the standard nerve-sparing plane — **Actually:** interfascial.
- **Trap:** transperitoneal prevents lymphocele — **Actually:** fluid still loculates.
- **Trap:** lymphocele usually presents as DVT — **Actually:** irritative voiding, pelvic pressure.
- **Trap:** obesity, large gland raise margins — **Actually:** tumour grade and stage do.
- **Trap:** early PDE5i improves later response — **Actually:** three RCTs negative.
- **Trap:** catheter out day 3 is safe — **Actually:** more retention; wait ≥5 days.
- **Trap:** Retzius-sparing is better overall — **Actually:** equal at 12 months, ↑ margins.
- **Trap:** post-TURP balloon marks bladder neck — **Actually:** falls into defect; cut too distal.
- **Trap:** extensive PLND improves survival — **Actually:** staging only; ↑ morbidity.